Mini-Project: Library Management System

Objective

Create a simple library management system that manages a collection of books. Each book should be represented as an object, and you will store these objects in an array.

Requirements

1. **Create a Book Class**:
    - Fields:
        - String title
        - String author
        - String isbn
        - boolean isAvailable (indicates if the book is available for borrowing)
    - Methods:
        - Constructor to initialize all fields.
        - Getters and setters for each field.
        - toString() method that returns a string representation of the book (title, author, ISBN, availability).

2. **Create a Library Class**:
    - Fields:
        - An array of Book objects (size can be determined by you).
    - Methods:
        - Constructor to initialize the array.
        - Method to add a book to the library.
        - Method to display all books in the library.
        - Method to borrow a book by its ISBN (set isAvailable to false).
        - Method to return a book by its ISBN (set isAvailable to true).
        - Method to search for a book by title or author.

3. **Main Class**:
    - Create an instance of the Library class.
    - Add some sample books to the library.
    - Implement a simple text-based menu that allows the user to:

- View all books.
- Borrow a book.
- Return a book.
- Search for a book by title or author.
- Exit the program.

Example Output

When the program runs, it should allow users to interact as follows:

Welcome to the Library Management System

1. View all books

2. Borrow a book

3. Return a book

4. Search for a book

5. Exit

Please enter your choice: 1

Implementation Steps

1. **Define the Book class.**
2. **Define the Library class.**
3. **Implement the main class to handle user interaction.**

Tips

- Use a loop in the main class to keep showing the menu until the user chooses to exit.
- Use conditionals to handle user choices.
- Ensure to handle cases where a book is not found when borrowing or returning.

Submission

- Submit the Java files for each class and a brief report (1-2 paragraphs) explaining your design choices and any challenges you encountered.